LUT University Summer 2026
BK10A6400 Basics of FE Analysis
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Bonus task 1
MATLAB
Possibility to earn five bonus points
Deadline: Before the lecture on 3rd of June

The aim of this bonus task is to practise the basic use of MATLAB and to connect simple matrix operations to a basic finite element problem. In particular, the task illustrates why an element stiffness matrix is singular before boundary conditions are applied, and how boundary conditions make the reduced stiffness matrix solvable.

Read the chapter "The First Few Steps" from the book *Programming for Computations – MATLAB/Octave* before doing the tasks.

Task 1: MATLAB basics (1 point)

Do Exercise 1.1 from the chapter "The First Few Steps".

Write the error messages in your report and explain briefly what they mean in these cases. Also describe how the errors can be corrected.

Task 2: Determinant and rank of a matrix (1 point)

Compute the determinant and rank of the matrix $\boldsymbol{A}$ presented in the code file `matrix_vector_product.m` from the chapter "The First Few Steps".

Write the values of the determinant and rank in your report. Based on the mathematics courses you have taken, explain what the determinant and rank tell about the matrix $\boldsymbol{A}$.

Hint: You can use the following MATLAB commands:

```
help det
help rank
```

Task 3: Stiffness matrix of a two-node axial rod element (3 points)

Write a MATLAB function that computes the local stiffness matrix of a two-node axial rod/bar element,

$$\overline{\boldsymbol{K}} = \frac{EA}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}. \tag{1}$$

Use the parameter values

$$E = 1, \qquad A = 1, \qquad L = 1.$$

These values are not physically realistic here. They are used only to keep the numerical values simple.

Compute the determinant, rank and eigenvalues of the full element stiffness matrix. Then apply a displacement boundary condition by eliminating either the first row and first column, or the second row and second column. Compute the determinant, rank and eigenvalues of the reduced stiffness matrix.

In your report, explain the following points:

- Why is the full element stiffness matrix singular before boundary conditions are applied?
- What is the physical meaning of the rank deficiency?

- What changes after one nodal displacement is fixed?
- Why can the reduced stiffness matrix be used to solve the unknown displacement?

MATLAB function

Write a function for the element stiffness matrix.

```
function K = Kloc(E,A,L)
%KLOC Local stiffness matrix of a two-node axial rod element.
%
% OUTPUT:
%   K = 2 x 2 local element stiffness matrix
%
% INPUT:
%   E = Young's modulus
%   A = cross-sectional area
%   L = element length

% Write the stiffness matrix here

end
```

Main program

Write a main program that calls the function and computes the requested quantities.

```
% Main program for Bonus task 1
clear; close all; clc;

% Parameters
E = 1;
A = 1;
L = 1;

% Element stiffness matrix
K = Kloc(E,A,L);

% Determinant, rank and eigenvalues of the full stiffness matrix
detK  = det(K);
rankK = rank(K);
eigK  = eig(K);

% Apply boundary condition by eliminating one row and one column
% Example: displacement of node 1 is fixed
Kred = K(2,2);

% Determinant, rank and eigenvalues of the reduced stiffness matrix
detKred  = det(Kred);
rankKred = rank(Kred);
eigKred  = eig(Kred);

% Print results
disp('Full stiffness matrix K:')
disp(K)
disp('det(K), rank(K), eig(K):')
disp(detK)
disp(rankK)
disp(eigK)

disp('Reduced stiffness matrix Kred:')
disp(Kred)
```

```
disp('det(Kred), rank(Kred), eig(Kred):')
disp(detKred)
disp(rankKred)
disp(eigKred)
```

Points

The maximum number of points is 5.

- Task 1: Correct error messages and explanations (1 point)
- Task 2: Correct determinant and rank of matrix $\boldsymbol{A}$, with interpretation (1 point)
- Task 3: Correct full stiffness matrix and its determinant, rank and eigenvalues (1 point)
- Task 3: Correct reduced stiffness matrix and its determinant, rank and eigenvalues after applying the boundary condition (1 point)
- Task 3: Clear MATLAB code and clear explanation of the physical meaning of the results (1 point)

Submission

Submit the following files in Moodle:

- A short report in PDF format
- The main MATLAB file, for example `main_bonus1.m`
- The MATLAB function file `Kloc.m`

The report must include the requested numerical results, short explanations and the main conclusions. The MATLAB files must be submitted in `.m` format. Do not submit `.mat` files.

Please note that points are given only if both the report and the MATLAB files are submitted.